Final Mock Exam — Set B — Solutions

Lectures 1–12 (comprehensive)

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Overview of the final mock exam — Set B

Problem	Topic	Lecture slides	Points
P1	Splines (beyond linearity)	Chapter 7 — Moving Beyond Linearity (Lecture 9)	16
P2	Generalised additive models	Chapter 7 — Moving Beyond Linearity (Lecture 9)	8
P3	Trees by hand	Chapter 8 — Tree-Based Methods (Lecture 10)	20
P4	Ensembles (bagging/RF/boosting)	Chapter 8 — Tree-Based Methods (Lecture 10)	16
P5	Neural networks by hand	Chapter 10 — Deep Learning (Lecture 11)	20
P6	Multiple testing	Chapter 13 — Multiple Testing (Lecture 12)	20
P7	Cumulative essentials	Chapters 2–6 (Lectures 1–8)	20
Σ			**120**

Exam conditions

120 minutes for 120 points ($\approx$ 1 point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). All required constants (e-powers and logs) are given in the problems. Justify every answer.

Problem 1 — task

Problem 1 (16 P) — Moving beyond linearity: splines

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

Throughout: regression splines on a single predictor X with $K = 5$ interior knots.

(a) **[4 P]** Degrees of freedom of a cubic spline with K interior knots — justify by counting parameters and constraints; evaluate for $K = 5$.

(b) **[4 P]** Degrees of freedom of a *natural* cubic spline with $K = 5$ knots; which extra constraint acts where, and why is it useful?

(c) **[4 P]** Smoothing spline $\sum_i (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$: role of the two terms; behaviour of $\hat{g}$ as $\lambda \to 0$ and $\lambda \to \infty$.

(d) **[4 P]** Truncated-power basis of a cubic spline with one knot ξ; define $(x - \xi)^3_+$; why does adding it keep the fit smooth at ξ?

Problem 1 — solution (a) and (b)

Solution P1 — (a) and (b)

(a) K knots $\Rightarrow$ $K+1$ cubic pieces $\times\, 4$ coefficients $= 4K+4$ parameters; continuity of g and g' and g'' at each knot $\Rightarrow$ $3K$ constraints:

$$\mathrm{df} = 4(K+1) - 3K = K+4 \qquad \Rightarrow \qquad K=5: \ \mathrm{df}=9.$$

(Equivalently: basis 1, x, x^2, x^3 plus one truncated power per knot, $4+K$.)

(b) Natural spline: *linear beyond the boundary knots* (regions $X < \xi_1$ and $X > \xi_K$); killing the quadratic and cubic terms there costs 2 parameters per boundary region, 4 in total:

$$\mathrm{df} = (K+4) - 4 = K \qquad \Rightarrow \qquad K=5: \ \mathrm{df}=5.$$

Why useful: splines are most erratic near the boundaries where data are sparse; the linearity constraint stabilises the fit there (lower variance for a little extra bias).

Problem 1 — solution (c) and (d)

Solution P1 — (c) and (d)

(c) RSS term = fidelity to the data; $\lambda \int g''(t)^2 dt$ penalises curvature (wiggliness); λ trades the two off.

- $\lambda \to 0$: no penalty $\Rightarrow$ $\hat{g}$ *interpolates* the training points exactly — extremely wiggly, zero training RSS, severe overfitting (effective df $\to n$).
- $\lambda \to \infty$: curvature infinitely expensive $\Rightarrow g'' \equiv 0 \Rightarrow \hat{g}$ is the *least-squares straight line* (effective df $\to 2$).

λ (chosen by LOOCV) steers the effective df smoothly between n and 2.

(d) $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$ with

$$(x-\xi)_+^3 = \begin{cases} (x-\xi)^3 & x > \xi \\ 0 & x \leq \xi \end{cases}$$

$(x-\xi)_+^3$ has value and first and second derivative 0 at $\xi \Rightarrow$ only the third derivative jumps: the fit stays continuous with continuous g' and g'' — a cubic spline.

Problem 2 — task

Problem 2 (8 P) — Generalised additive models

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

(a) **[3 P]** General form of a GAM for quantitative Y; how does it generalise multiple linear regression?

(b) **[3 P]** One interpretability advantage concerning the f_j; the main structural limitation — and its repair.

(c) **[2 P]** Backfitting in one or two sentences.

Problem 2 — solution

Solution P2 — GAMs

(a) $Y = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p) + \varepsilon$.
Linear regression is the special case $f_j(X_j) = \beta_j X_j$; a GAM replaces each linear term by a smooth non-linear f_j (spline, local regression) but keeps the *additive* structure.

(b) *Advantage:* additivity $\Rightarrow$ the effect of each X_j is its own function f_j, which can be plotted and interpreted individually while the other predictors are held fixed — one panel per predictor, as readable as a coefficient table.
Limitation: no *interactions* — the effect of X_j cannot depend on X_k.
Repair: add interaction components manually, e.g. $X_j \times X_k$ or a two-dimensional smooth $f_{jk}(X_j, X_k)$.

(c) *Backfitting:* cycle through $j = 1, \ldots, p$ and re-fit each f_j to the partial residuals $r_i = y_i - \beta_0 - \sum_{k \neq j} f_k(x_{ik})$ with all other functions held fixed; repeat until convergence.

Problem 3 (20 P) — Trees by hand

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

Delivery vans x and parcels/day y (1,000s) for $n = 6$ depots; one split at s: $R_1 = \{x < s\}$ and $R_2 = \{x \geq s\}$ with region-mean predictions.

i	1	2	3	4	5	6
x_i	10	12	14	16	18	20
y_i	6	4	5	11	13	12

(a) **[8 P]** Evaluate the cutpoints $s = 15$ and $s = 17$ by total RSS; which is chosen? (b) **[2 P]** Prediction for a new depot with $x = 13$. (c) **[6 P]** Node with 12 obs: 6 "Premium" / 3 "Standard" / 3 "Basic": proportions; Gini; entropy ($\ln 0.5 = -0.6931$; $\ln 0.25 = -1.3863$); predicted class. (d) **[4 P]** Cost-complexity pruning: role of α in $\sum_m \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$.

Problem 3 — solution (a) and (b)

Solution P3 — (a) and (b)

(a) Region means and RSS per cutpoint ($\bar{y} = 51/6 = 8.5$, root RSS $= 77.5$):

s	R_1 (y-values)	$\hat{y}_{R_1}$	R_2 (y-values)	$\hat{y}_{R_2}$	total RSS
15	{6,4,5}	5	{11,13,12}	12	$2 + 2 = 4$
17	{6,4,5,11}	6.5	{13,12}	12.5	$29 + 0.5 = 29.5$

Details: $s = 15$: $(6-5)^2 + (4-5)^2 + (5-5)^2 = 2$; $(11-12)^2 + (13-12)^2 + (12-12)^2 = 2$.
$s = 17$: $(6-6.5)^2 + (4-6.5)^2 + (5-6.5)^2 + (11-6.5)^2 = 0.25 + 6.25 + 2.25 + 20.25 = 29$; $0.25 + 0.25 = 0.5$.
Greedy recursive binary splitting picks the smaller RSS: s $= 15$ ($4 \ll 29.5$) — it separates the three low-throughput depots (mean 5) from the three high ones (mean 12).

(b) $x = 13 < 15 \Rightarrow$ region $R_1 \Rightarrow \hat{y} = \hat{y}_{R_1} = 5$ (i.e. 5,000 parcels per day).

Problem 3 — solution (c) and (d)

Solution P3 — (c) and (d)

(c) $\hat{p}_{\text{Prem}} = 6/12 = 0.5$; $\hat{p}_{\text{Std}} = \hat{p}_{\text{Basic}} = 3/12 = 0.25$.

$$G = 0.5(1-0.5) + 0.25(1-0.25) + 0.25(1-0.25) = 0.25 + 0.1875 + 0.1875 = 0.625$$

$$D = -(0.5\ln 0.5 + 0.25\ln 0.25 + 0.25\ln 0.25) = 0.5 \cdot 0.6931 + 0.5 \cdot 1.3863 = 0.3466 + 0.6931 = 1.040$$

Predicted class: majority = **Premium** (estimated probability 0.5). Both measures are 0 for a pure node — this node is quite impure (three classes present).

(d) α = price per terminal node $|T|$:

- $\alpha = 0$: criterion = training RSS $\Rightarrow$ full tree T_0 is optimal.
- $\alpha \uparrow$: leaves must pay for themselves; branches with small RSS gain are pruned $\Rightarrow$ nested sequence of ever smaller subtrees (weakest-link pruning).
- Choice: *cross-validation* over α; refit the chosen subtree on all data — balances bias against variance.

Problem 4 (16 P) — Ensembles

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

(a) **[6 P]** B identically distributed trees with variance σ^2 and pairwise correlation ρ: $\mathrm{Var}\left(\frac{1}{B}\sum_b \hat{f}_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. (i) Why does this show variance reduction and what limits it? (ii) Evaluate for $B = 500$ and $\rho = 0.7$ and $\sigma^2 = 4$; compare with one tree. (iii) Limit as $B \to \infty$?

(b) **[4 P]** Random forests: modification at each split; recommended m for classification with $p = 36$; why does it improve on bagging (use the formula)?

(c) **[4 P]** Boosting: the three tuning parameters; the slow-learning idea.

(d) **[2 P]** What does the OOB error estimate and why is it free?

Problem 4 — solution (a) and (b)

Solution P4 — (a) and (b)

(a) (i) Averaging drives the second term $\frac{1-\rho}{B}\sigma^2 \to 0$ (the independent part of the trees' variability); the first term $\rho\sigma^2$ is independent of B — correlation between bootstrap trees limits the reduction.
(ii) $\text{Var} = 0.7 \cdot 4 + \frac{1-0.7}{500} \cdot 4 = 2.8 + 0.0024 = 2.802$ vs. $\sigma^2 = 4$ for a single tree ($\approx 30\%$ less).
(iii) $B \to \infty$: variance $\to \rho\sigma^2 = 2.8$ (floor; more trees never overfit, they only stabilise).

(b) At *each split* only a random subset of m of the p predictors may be used; classification default $m \approx \sqrt{p}$ $\Rightarrow p = 36$: $m = \sqrt{36} = 6$.
With bagging a dominant predictor tops nearly every tree $\Rightarrow$ trees alike $\Rightarrow$ ρ high. Restricting candidates *decorrelates* the trees ($\rho \downarrow$), which lowers the floor $\rho\sigma^2$ — the part of the variance that averaging alone cannot remove.

Problem 4 — solution (c) and (d)

Solution P4 — (c) and (d)

(c) Boosting for regression trees:

- B — **number of trees:** added sequentially; too many $\Rightarrow$ slow overfitting; choose by CV.
- λ — **shrinkage** (e.g. 0.01 or 0.001): scales each new tree's contribution; smaller $\lambda \Rightarrow$ slower learning $\Rightarrow$ larger B needed.
- d — **splits per tree** (interaction depth): complexity and interaction order of each tree; often stumps ($d = 1$) suffice (additive model).

Slow learning: each tree is fitted to the current *residuals* and only the fraction λ is added — the model improves gradually exactly where it still errs, instead of fitting the data hard at once; this tends to generalise better.

(d) Each bootstrap sample misses $\approx 1/3$ of the observations (out-of-bag); predicting every observation only with trees that never saw it yields the **OOB error** — a valid *test-error* estimate, free as a by-product of the bootstrap (no validation set or CV loop needed).

Problem 5 (20 P) — Neural networks by hand

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Input $x = (2,1)$; hidden ReLU units $g(z) = \max\{0,z\}$; linear output:

$$A_1 = g(-2 + 3x_1 - 1x_2) \qquad A_2 = g(1 - 2x_1 + 1x_2) \qquad f(x) = -1 + 2A_1 + 3A_2$$

(a) **[8 P]** Forward pass step by step: pre-activations; A_1 and A_2; output $f(x)$. What did the ReLU do to unit 2 and why are non-linear activations essential?

(b) **[6 P]** Logits $z = (2.0,\ 1.0,\ -1.0)$; true class 1: all three softmax probabilities and the cross-entropy loss ($e^2 = 7.389$; $e^1 = 2.718$; $e^{-1} = 0.368$; $\ln 10.475 = 2.349$).

(c) **[4 P]** Parameter count of a $p = 6 \to k = 4 \to K = 3$ network with biases (show both layers).

(d) **[2 P]** Conv layer: 32×32 input; $F = 5$; $P = 0$; $S = 1$: output size via $(W - F + 2P)/S + 1$.

Problem 5 — solution (a) forward pass

Solution P5 — (a) forward pass

Pre-activations:

$$z_1 = -2 + 3 \cdot 2 - 1 \cdot 1 = 3 \qquad z_2 = 1 - 2 \cdot 2 + 1 \cdot 1 = -2$$

ReLU activations:

$$A_1 = \max\{0,3\} = 3 \qquad A_2 = \max\{0,-2\} = 0$$

Output:

$$f(x) = -1 + 2 \cdot 3 + 3 \cdot 0 = 5$$

The ReLU *switched unit 2 off*: its negative pre-activation is clipped to 0, so it contributes nothing for this input (though it can for others).

Why essential: with a linear g the whole network collapses into one linear function of x regardless of depth — the non-linearity is what lets hidden layers build flexible functions (piecewise-linear surfaces with ReLU).

Solution P5 — (b) softmax and cross-entropy

Exponentials and their sum:

$$e^{2.0} = 7.389 \qquad e^{1.0} = 2.718 \qquad e^{-1.0} = 0.368 \qquad \Rightarrow \qquad 7.389 + 2.718 + 0.368 = 10.475$$

Softmax probabilities:

$$\hat{p}_1 = \frac{7.389}{10.475} = 0.705 \qquad \hat{p}_2 = \frac{2.718}{10.475} = 0.259 \qquad \hat{p}_3 = \frac{0.368}{10.475} = 0.035$$

(sum $= 0.999 \approx 1$ ✓). Cross-entropy for true class 1:

$$-\ln \hat{p}_1 = -\ln \frac{e^2}{10.475} = -(2 - \ln 10.475) = 2.349 - 2 = 0.349$$

Sanity checks: a perfect prediction ($\hat{p}_1 = 1$) gives loss 0; predicting the uniform $1/3$ would give $\ln 3 \approx 1.099$ — here the network is right and fairly confident.

Problem 5 — solution (c) and (d)

Solution P5 — (c) parameters and (d) convolution

(c) Fully connected $6 \to 4 \to 3$ with biases:

Layer	Weights	Biases	Parameters
input $\to$ hidden	$6 \cdot 4 = 24$	4	28
hidden $\to$ output	$4 \cdot 3 = 12$	3	15
total			43

(d) Output width $= \dfrac{W - F + 2P}{S} + 1 = \dfrac{32 - 5 + 2 \cdot 0}{1} + 1 = 27 + 1 = 28 \Rightarrow$ output feature map 28×28: with no padding ($P = 0$) a 5×5 filter at stride 1 shrinks each spatial dimension by $F - 1 = 4$.

Problem 6 (20 P) — Multiple testing

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

(a) **[4 P]** $m = 10$ independent tests at $\alpha = 0.05$; all nulls true: define and compute the FWER ($(0.95)^{10} = 0.5987$); interpret.

(b) **[7 P]** Six ordered p-values 0.002; 0.009; 0.012; 0.015; 0.040; 0.300 — FWER control at $\alpha = 0.05$ with (i) Bonferroni and (ii) Holm; thresholds; rejected hypotheses; why is Holm never weaker than Bonferroni?

(c) **[6 P]** Benjamini–Hochberg at $q = 0.05$ on the same p-values: thresholds jq/m; rejections; what exactly does FDR control guarantee?

(d) **[3 P]** Confirmatory Phase-III trial with $m = 4$ pre-specified endpoints (one hit could support approval): FWER or FDR? Justify.

Problem 6 — solution (a) FWER

Solution P6 — (a) FWER for 10 independent tests

FWER $= \Pr(V \geq 1)$ — the probability of *at least one* false rejection among the m tests (V = number of falsely rejected true nulls).

With all 10 nulls true and independent tests at level 0.05:

$$\text{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (1 - 0.05)^{10} = 1 - (0.95)^{10} = 1 - 0.5987 = 0.4013$$

Interpretation: although each test individually errs with probability 5% only, the chance of at least one spurious "discovery" among 10 unadjusted tests is about 40% — multiple testing without adjustment makes false findings very likely.

Problem 6 — solution (b) Bonferroni and Holm

Solution P6 — (b) Bonferroni and Holm at level 0.05

Bonferroni: reject iff $p_{(j)} \leq \alpha/m = 0.05/6 = 0.00833$: only 0.002 passes ($0.009 > 0.00833$) $\Rightarrow$ reject $H_{(1)}$: **1 rejection**. (Note $0.009 < 0.05$ yet Bonferroni does not reject it.)

Holm (step-down): compare $p_{(j)}$ with $\alpha/(m-j+1)$; stop at first failure:

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.009	0.012	0.015	0.040	0.300
$\alpha/(m-j+1)$	0.00833	0.0100	0.0125	0.0167	0.0250	0.0500
decision	✓	✓	✓	✓	**stop**	—

$0.015 \leq 0.0167$ but $0.040 > 0.0250 \Rightarrow$ reject $H_{(1)}$ to $H_{(4)}$: **4 rejections**.

Holm $\supseteq$ Bonferroni: Holm's hurdles $\alpha/(m-j+1) \geq \alpha/m$ for all j (equality only at $j=1$) $\Rightarrow$ uniformly more powerful — and it still controls the FWER without independence assumptions.

Problem 6 — solution (c) and (d)

Solution P6 — (c) Benjamini–Hochberg and (d) FWER vs FDR

(c) Thresholds $jq/m = j \cdot 0.05/6$:

j	1	2	3	4	5	6
$p_{(j)}$	0.002	0.009	0.012	0.015	0.040	0.300
jq/m	0.00833	0.0167	0.0250	0.0333	0.0417	0.0500
$p_{(j)} \leq jq/m$?	yes	yes	yes	yes	**yes**	no

Largest such j: $L = 5$ ($0.040 \leq 0.0417$; note $0.300 > 0.0500$) $\Rightarrow$ reject *all* of $H_{(1)}$ to $H_{(5)}$: **5 rejections**. *Guarantee:* FDR $= E[V/\max(R,1)] \leq q$ — *on average* at most 5% of rejected hypotheses are false discoveries (no bound on $\Pr(V \geq 1)$; that is FWER).

(d) Control the **FWER**. A *single* false positive is very costly (could support approving an ineffective drug), so guard against *any* Type I error, not just their average proportion. With only $m = 4$ endpoints an FWER procedure (Holm/Bonferroni at $0.05/4 = 0.0125$) loses little power; FDR is too permissive for a confirmatory trial.

Problem 7 — task

Problem 7 (20 P) — Cumulative essentials

Lecture slides: Chapters 2–6 (Lectures 1–8).

(a) **[4 P]** Screening confusion matrix (TP=40, FN=10, FP=30, TN=120): error rate, sensitivity, specificity, precision; which number traces the ROC curve and how do sensitivity/specificity move?

(b) **[4 P]** KNN: (i) predict at $x_0 = 5$ with $k = 3$ from (1,3), (4,6), (6,8), (7,7), (10,12); (ii) how k drives the bias–variance trade-off; most flexible extreme; why is 1-NN training error 0?

(c) **[4 P]** k-fold CV: what it does; one reason over LOOCV; one over a single validation split.

(d) **[2 P]** $p = 2{,}000$ predictors, only a handful matter: ridge or lasso? Where does PCR sit?

(e) **[6 P]** Match {linear regression; lasso; logistic regression; random forest} — each once: (i) \$/schooling-year with CI; (ii) BP from 8,000 genes (n=150) with a short gene list; (iii) spam yes/no with odds for auditors; (iv) machine failure from interacting sensors, accuracy only.

Problem 7 — solution (a) and (b)

Solution P7 — (a) confusion matrix and (b) KNN

(a) 50 diseased, 150 healthy, $n = 200$:

$$\text{err} = \frac{30+10}{200} = 0.200 \quad \text{sens} = \frac{40}{50} = 0.800 \quad \text{spec} = \frac{120}{150} = 0.800 \quad \text{prec} = \frac{40}{70} = 0.571$$

Varying the *classification threshold* traces the ROC curve: lowering it raises sensitivity (fewer misses) but lowers specificity (more false alarms); raising it reverses.

(b) (i) Distances from $x_0 = 5$: $4, 1, 1, 2, 5$; nearest three $x = 4, 6, 7$ with $y = 6, 8, 7 \Rightarrow \hat{y}(5) = \frac{6+8+7}{3} = 7$.
(ii) Small k = flexible: low bias, high variance; large k = smooth: high bias, low variance. Most flexible extreme = k = 1 (flexibility $\propto 1/k$). The training error of 1-NN is 0 because each point's nearest neighbour is *itself*.

Problem 7 — solution (c) and (d)

Solution P7 — (c) cross-validation and (d) lasso vs ridge/PCR

(c) k-fold CV splits the data into k equal folds; each fold serves once as validation while the other $k-1$ train the model, and the k errors are averaged.

- *vs. LOOCV:* the n leave-one-out fits are nearly identical $\Rightarrow$ their errors are highly correlated $\Rightarrow$ the LOOCV average has *higher variance* (and costs n fits).
- *vs. a single split:* one split is highly variable and biased upward (only part of the data trains); k-fold trains on $(k-1)/k$ and averages k estimates $\Rightarrow$ less bias, more stability.

(d) Lasso: the ℓ_1 penalty sets coefficients *exactly to zero* $\Rightarrow$ variable selection; ideal for a sparse truth (a handful of signals among 2,000) and an interpretable model. Ridge keeps all 2,000; PCR is likewise *not* sparse (regresses on a few components, all variables load) — neither delivers lasso's short list.

Problem 7 — solution (e) matching

Solution P7 — (e) method matching (each once)

(i) **Linear regression** — quantitative response with $n \gg p$; goal is *inference*: an effect size with a confidence interval from OLS standard errors.

(ii) **Lasso** — $p = 8{,}000 \gg n = 150$: OLS infeasible; the ℓ_1 penalty regularises and returns a short list of relevant markers.

(iii) **Logistic regression** — binary response; coefficients give odds ratios $e^{\hat{\beta}_j}$: interpretable for auditors.

(iv) **Random forest** — automatic non-linearities and interactions; strong accuracy with many mixed predictors; interpretability not required.

One-line rule of thumb

Inference on few predictors $\to$ linear/logistic regression; sparse high-dimensional selection $\to$ lasso; pure predictive accuracy with complex structure $\to$ random forest.